

Bond degradation of basalt fiber-reinforced polymer (BFRP) bars exposed to accelerated aging conditions



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HIGHLIGHTS

- Durability of conditioned basalt and glass FRP bars was investigated.
- Parameters included the exposure type and duration.
- Basalt bars showed excellent adhesion and bond strengths to concrete.
- Conditioning reduced the bond strength of basalt bars by 14–25%.
- BPE model was calibrated to consider the bar conditioning.

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ABSTRACT

Bond durability of sand-coated basalt fiber-reinforced polymer (BFRP) bars was investigated. Pullout specimens were tested under direct tensile load after being exposed to accelerated conditioning environments. The test parameters included the bar material (basalt and glass), the conditioning environment (acid, saline, and alkaline), and the duration of exposure (30, 60, and 90 days). The bond behavior of the tested specimens was reported in terms of stress-slip response, bond strength, bar slip, adhesion, and failure mechanism. The BFRP bars showed higher adhesion and bond strengths to concrete than the ribbed glass fiber-reinforced polymer (GFRP) bars irrespective of the fiber type and the exposure condition. The absorption of the bar material and its quality of manufacturing affected the bond behavior of the conditioned specimens. All specimens failed in pullout mode by inter-laminar shear between the layers of the bar. Conditioning reduced the bond strength of the BFRP bars by 14–25% of their initial strength. Ribbed GFRP specimens subjected to acid exposure suffered the highest loss in bond strength (25%) compared to 17% for GFRP specimens exposed to alkaline and ocean water. Conditioning had insignificant influence on the slip of the bars.

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1. Introduction and background

Recently, basalt fiber-reinforced polymer (BFRP) bars have emerged as a promising alternative to conventional glass fiber-reinforced polymer (GFRP) materials. However, North American codes do not recognize the BFRP bars as reinforcing materials for concrete structures. The few studies conducted to date on BFRP bars [12,13,5,16] do not provide confidence to potential users of the material. As a result, there is lack of knowledge on the performance of the bars in concrete, in particular their durability while in service.

One of the major aspects towards the wide acceptance of BFRP bars in construction is their bond durability since bond dictates the integrity and ductility of the concrete member. It is well established that the fiber type used in the manufacturing of FRP bars greatly affects their bond durability when embedded in concrete [8,15]. If the fibers fail to withstand the surrounding exposures during service, the FRP bar should not be recommended as reinforcement. In fact, both the fiber and the resin influence the mechanical properties of any FRP bar and hence its bond performance to concrete [7,10]. On the other hand, exposing FRP bars to aqueous solutions, chemical attacks, or combinations of severe environments may result in degradation of the mechanical and physical properties of the bars [2] and hence bond deterioration. Therefore, bond investigation of the newly developed BFRP bars is a vital step towards their recognition as reinforcing materials.

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Bond degradation mechanism is a complex process. Moisture can diffuse into the resin and between the resin and fibers leading to cracks and voids in the bar material. As a result, the rate of diffusion of water and solution molecules increases causing the deterioration of the interfacial surface of the bars and the degradation of bond to the surrounding concrete. This effect is more predominant if the moisture contains chloride ions as usually found in oceanic environments [6,13]. When exposed to alkaline or acid environments, the FRP material degrades because of the chemical reaction that takes place between the fibers and the attacking solution [19]. This could be prevented if appropriate resin protection is provided to the fibers.

Several studies have reported on the chemical attack of FRP bars and their effect on their bond degradation. Chen et al. [7] investigated the durability of glass and carbon FRP (GFRP and CFRP, respectively) bare and embedded bars in concrete. Five solutions, namely tap water, two alkaline solutions simulating normal and high strength concrete environments, saline solution, and mix of alkaline and chloride solutions were used. The specimens were also exposed to wet-and-dry and freeze-and-thaw cycles before being tested. Significant strength reductions resulting from the accelerated exposures were reported for both bare and embedded bars. The authors concluded that continuous immersion in chemical attacks resulted in higher bond deterioration than the exposure to wet-and-dry and freeze-and-thaw cycles.

In another study, Davalos et al. [10] reported on the surface degradation of GFRP and CFRP bars embedded in high strength concrete. Forty-eight pullout specimens were exposed to three environmental conditions: tap water at normal temperature, tap water at 60 °C, and thermal cycles between 20 °C and 60 °C. The authors reported reduction of 4–10% in the bond strength of CFRP bars and 0–20% for GFRP bars. It was concluded that the deterioration of FRP bars during conditioning was the main reason for bond degradation. It was also found that conditioning increased the free-end slip prior to reaching the bond strength of the bar.

Robert and Benmokrane [17] studied the effect of tap water at different temperatures (23, 40, and 50 °C) for three exposure durations (60, 120, and 180 days) on the bond strength of sand-coated GFRP bars. The study concluded that the bond strength of the bars decreased with increasing the immersion duration whereas the temperature exposure had insignificant influence on the test results. The authors also concluded that the bond strength of the GFRP bars was mainly governed by their outer surface treatment.

Belarbi and Wang [4] tested thirty-six pullout specimens cast with plain and fiber-reinforced concrete and reinforced with GFRP and CFRP bars. The specimens were exposed to accelerated aging conditions consisting of successive exposures to saline solution, freeze-and-thaw cycles, and high temperature cycles. It was found that the rate and severity of bond degradation were more pronounced in the GFRP specimens than in the CFRP specimens. Reduction of 28% was reported for the bond strength in plain concrete compared to 6% in fiber-reinforced concrete.

Zhou et al. [20] studied the effect of different environmental conditions on the bond strength of ninety pullout concrete cubes reinforced with 17 mm diameter ribbed GFRP bars. Specimens reinforced with steel bars were also tested for comparison. Exposure conditions included tap water, alkaline solution (pH = 13.5), ocean water, and acid solution (pH = 2) for 30, 60, and 90 days at 20 °C. No sign of bond degradation in the GFRP bars was reported except in those specimens immersed in the acid solution.

It is important to note that the conducted literature review revealed large discrepancies in the reported bond results. This can be attributed to the different test methods that were adopted and to the wide variety in the characteristics of the tested bars. On the other hand, very few studies were devoted to investigate the bond of BFRP bars, not to mention their durability performance.

Therefore, the current paper aims at narrowing the gap in our knowledge on such performance. The work presented herein consists a part of a large research study on the use of BFRP bars as reinforcement. The testing program is formulated based on the results of an earlier experimental investigation on the bond durability of two types of BFRP bars and one GFRP bar [13]. In the current paper, further aging conditions are considered along with the effect of exposure duration on the bond performance of the bars. The work includes the calibration of the analytical model developed by Eligehausen et al. [14], which describes the ascending branch of the bond stress–slip curve to account for the degradation of the bars after conditioning.

2. Experimental program

The experimental program consisted of sixty-two pullout specimens prepared according to the ACI 440.3R-04 [1] provisions. Concrete cubes of 200 mm side were concentrically reinforced with 1000 mm long BFRP and GFRP bars as shown in Fig. 1a. An embedment length of $5d_b$, where d_b is the bar diameter, was selected to ensure a pullout mode of failure. Plastic tubes were placed at the top of the embedded portion of the bar to act as bond breakers. Specimens were exposed to three different solutions, each simulating a specific field exposure, for three different durations. The following describes the properties of the bars, the concrete mix, the conditioning exposures, and the test setup.

2.1. Properties of BFRP and GFRP bars

BFRP and GFRP bars with nominal diameter of 12 mm were used in this study. Actual diameters were measured by volume displacement as recommended by ACI 440.3R-04 [1]. Table 1 lists the mechanical properties of the bars as provided by the manufacturers and as tested by the authors. The surface treatment of each bar type is shown in Fig. 2. BFRP bars were sand-coated with shallow spiral indentations spaced at 2.75 mm on their surface. GFRP bars had a ribbed surface of 0.5 mm height and spaced at 5.5 mm along the bar. The authors are aware that different surface treatments of FRP bars lead to different bond test results. It is therefore important to note that the aim of this paper is to assess the bond performance of both types of bars rather than comparing their bond performances.

2.2. Concrete mix

Concrete mix of compressive cubic strength 60 MPa was used to cast the pullout specimens. This high strength concrete was used to prevent splitting of concrete prior to developing the bond strength of the FRP bars. Table 2 shows the proportions of the concrete mix used in casting the tested specimens. After 28 days of curing in large water tanks, the specimens were left outside the laboratory before being conditioned.

2.3. Environmental conditioning

Three accelerated aging environments were considered in this study. These were (1) acid solution with pH = 2.0, simulating polluted industrial regions and prepared from a mix of hydrochloric acid (HCl) and sulfuric acid (H_2SO_4) in the ratio of 2:1, respectively; (2) ocean water, simulating the climate of coastal areas and prepared from a mix of sodium sulfate (Na_2SO_4) and sodium chloride (NaCl) in the ratio of 5:30 g/L, respectively; and (3) alkaline solution with pH = 12.5, simulating the alkaline environment in normal reinforced concrete and prepared from a mix of calcium hydroxide ($Ca(OH)_2$), sodium hydroxide (NaOH), and potassium hydroxide (KOH) in the ratio of 2:2.4:19.6 g/L, respectively. For each exposure, eighteen specimens (nine BFRP bars and nine GFRP bars) were conditioned for 30, 60, and 90 days before being tested. Three temperature-controlled steel tanks were fabricated and filled with the conditioning solutions. The temperature inside the tanks was maintained at 60 °C during exposure. The tanks were equipped by electrical pumps to ensure uniform distribution of the solutions around the immersed specimens.

2.4. Test matrix

The test matrix is shown in Table 3. Replicate specimens were tested to ensure the reliability of the test results. All specimens are labeled in the following order: the bar type (B for basalt and G for glass FRP bars), the conditioning exposure (A for acid, K for alkaline, S for ocean water, and C for unconditioned control specimens), the exposure duration in days (30, 60, and 90), and the specimen's number in each set. For instance, specimen BK-60-2 refers to a specimen reinforced with BFRP bar (B) and exposed to alkaline solution (K) for 60 days (60). The digit (2) refers to the second specimen in its set. It is worth mentioning that, in order to

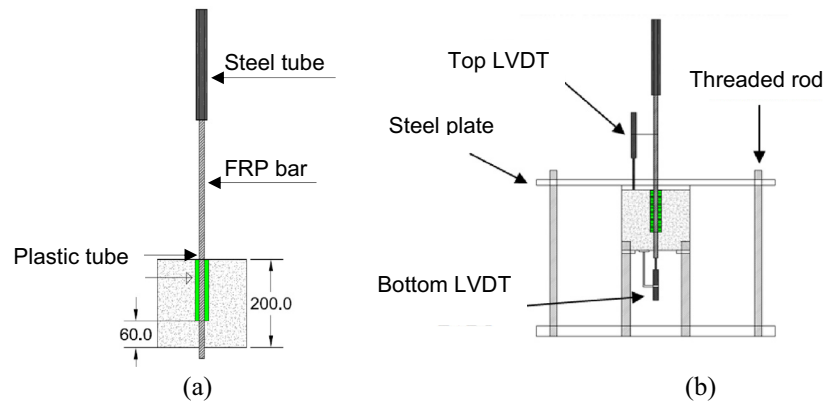


Fig. 1. Schematic of the (a) pullout specimen and (b) test setup.

Table 1
Mechanical properties of fiber-reinforced polymer bars.

Type	Diameter (mm)	Ultimate tensile strength (MPa)	Elongation at break (%)	Elastic modulus (GPa)	Moisture absorption (%)
BFRP	12 (12.3)	1168 (1017)	2.2 (2.12)	50 (48)	0.47
GFRP	12 (11.6)	1000 (964)	2.0 (2.10)	50 (46)	0.024

Note: Values between brackets are those tested by the authors.

Table 2
Constituents of the concrete mix used in this study.

Mix constituents	Quantity (kg/m ³)
Portland cement	330
Water	149
Coarse aggregate (20 mm)	600
Coarse aggregate (10 mm)	400
Fine aggregate (5 mm)	600
Fine aggregate (dune sand)	365
Plasticizer	6

monitor the variation in the concrete compressive strength after conditioning, concrete cubes (150 × 150 mm) were cast and subjected to similar curing, conditioning, and exposure duration as the pullout specimens.

2.5. Pullout test setup

Pullout tests were carried out in an MTS Universal Testing Machine of a tensile capacity of 1000 kN. The free portion of the FRP bar was gripped by hollow steel tube 350 mm long and 30 mm external diameter. The tubes were adhered to the FRP bar using a commercial epoxy. A pullout test rig was manufactured to attach the specimens in the testing machine. A schematic of the test setup is shown in Fig. 1b. Two linear variable displacement transducers (LVDT) were attached to the loaded end of the bar to record the slip of the bar relative to the top surface of concrete. A third LVDT measured the slip at the unloaded end of the bar that extended outside the concrete cube. The pullout force was applied at a rate of 1.2 mm/min. A data acquisition system recorded the load and slip readings at a rate of 10 readings/s.

3. Test results

The bond test results are shown in Table 3 in terms of maximum stress, slip corresponding to the maximum stress, bond stress at onset of slip, and residual stress at both loaded and unloaded ends. For all specimens, the bond strength was calculated as follows:

$$\tau = \frac{P_{\max}}{\pi d_b L_d}$$

where $P_{\max}$ is the maximum applied load, d_b is the bar diameter, and L_d is the bar embedment length. The bar slip at the unloaded end

was directly measured using the LVDT attached at the bottom of the specimen. At the loaded end, the bar slip was determined by subtracting the bar elongation from the reading of the upper LVDTs. The actual bar diameters were used in bond stress calculations.

3.1. Concrete compressive strength

Table 4 lists the variation in the concrete compressive strength after conditioning. It was observed that each exposure condition and duration affected the concrete strength differently. All conditioned cubes experienced an initial increase in their compressive strength after 30 days of immersion. This was attributed to the long curing period of concrete during its immersion in aqueous solutions. Concrete cubes immersed in the acid solution exhibited significant deterioration in their compressive strength after 60 and 90 days as a result of the acid attack to concrete, whereas their counterpart specimens immersed in alkaline solutions and ocean water showed insignificant variation in their compressive strengths.

3.2. Bond stress–slip response

Bond–slip relationships at the loaded and unloaded ends are depicted in Fig. 3 to Fig. 5 for representative BFRP and GFRP specimens. For all specimens, the ascending branches of the bond–slip curves showed a linear relationship, with larger slip encountered at the loaded end than in the unloaded ends at all stages of loading.

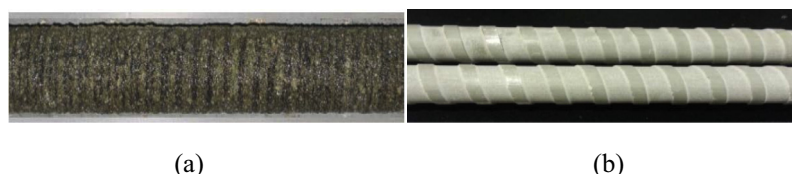


Fig. 2. (a) Basalt and (b) glass fiber-reinforced polymer bars used in this study.

Table 3

Bond test results of BFRP and GFRP specimens exposed to different solutions.

Specimen	Failure mode	$\tau_{\max}$ MPa	$\tau_{\max}/\sqrt{f'_c}$ $\sqrt{\text{MPa}}$	$s_{\max, fe}$ mm	$s_{\max, le}$ mm	$\tau_{\text{ons}, le}$ MPa	$\tau_{\text{ons}, fe}$ MPa	$\tau_{r, fe}$ MPa	$\tau_{r, le}$ MPa
<i>Unconditioned (control) specimens: specimens C</i>									
BC-1	P	26.39	3.41	N/A	N/A	N/A	N/A	N/A	N/A
BC-2	P	25.8	3.33	0.2	2.27	0.5	16.24	0	0
BC-3	P	28.48	3.68	N/A	N/A	N/A	N/A	N/A	N/A
BC-4	P	28.02	3.62	0.11	2.01	0.91	18.4	0	0
BC-5	P	30.51	3.94	N/A	2.56	0.53	N/A	N/A	0
GC-1	P	18.89	2.44	0.1	1.9	0.36	11.70	16.04	16.67
GC-2	P	18.92	2.44	0.05	1.71	0.34	11.98	16.4	16.05
GC-3	P	17.44	2.25	N/A	1.59	0.53	N/A	N/A	18.02
<i>Specimens conditioned in acid solution: specimens A</i>									
BA-30-1	P	23.66	2.75	0.25	2.43	2.58	14.67	0	0
BA-30-2	P	21.31	2.48	0.12	1.46	0.56	15.50	0	0
BA-30-3	P	24.69	2.87	0.02	1.96	2.6	15.60	0	0
GA-30-1	P	20.89	2.43	0.53	1.24	1.05	6.77	18.50	18.1
GA-30-2	S	18.47	2.15	0.00	1.97	0.54	2.05	17.60	17.6
GA-30-3	P	20.65	2.40	0.36	0.66	1.04	5.52	19.20	19.6
BA-60-1	P	24.70	3.07	0.00	2.46	0.28	24.70	0	0
BA-60-2	P	17.67	2.20	0.07	1.29	1.22	12.90	4.12	4.12
BA-60-3	P	26.39	3.29	0.68	2.12	0.67	0.73	0	0
GA-60-1	P	17.84	2.22	0.54	1.24	0.73	0.81	14.70	14.9
GA-60-2	S	17.35	2.16	N/A	1.14	0.51	N/A	N/A	16.3
GA-60-3	P	15.82	1.97	0.38	0.90	0.19	2.77	17.60	17.7
BA-90-1	P	24.20	3.26	0.00	2.24	0.67	20.95	0	0
BA-90-2	P	23.03	3.11	0.41	1.87	1.38	1.76	0	0
BA-90-3	P	21.00	2.83	0.49	2.74	0.22	0.69	0	0
GA-90-1	S	13.33	1.80	0.32	0.60	0.17	1.20	0	0
GA-90-2	P	18.52	2.50	N/A	0.58	3.78	N/A	N/A	16.5
GA-90-3	S	13.74	1.85	0.52	1.11	0.61	0.00	0	0
<i>Specimens conditioned in alkaline solution: specimens K</i>									
BK-30-1	P	24.79	2.96	0.11	3.20	0	N/A	0.00	0.00
BK-30-2	P	25.43	3.04	0.06	1.93	2.88	N/A	0.00	0.00
BK-30-3	P	28.40	3.39	0.14	4.00	0.94	5.00	0.00	0.00
GK-30-1	P	19.02	2.27	0.24	1.36	0.34	8.52	18.7	18.7
GK-30-2	P	18.32	2.19	0.24	1.71	0.67	10.04	17.5	17.5
GK-30-3	P	18.90	2.26	0.47	0.55	3.88	8.99	19.3	19.3
BK-60-1	P	25.77	2.81	0.14	2.09	0.28	10.50	0.00	0.00
BK-60-2	P	28.05	3.06	N/A	2.25	0.00	N/A	N/A	0.00
BK-60-3	P	25.77	2.81	0.13	2.28	0.56	6.40	0.00	0.00
GK-60-1	P	17.65	1.92	0.80	1.00	0.42	8.10	16.6	16.6
GK-60-2	P	18.65	2.03	0.21	0.62	0.49	10.80	16.7	16.7
GK-60-3	P	18.61	2.03	0.18	1.28	0.62	10.60	16.8	16.8
BK-90-1	P	26.39	2.91	0.11	2.31	0.56	N/A	N/A	0.00
BK-90-2	P	22.89	2.53	0.00	2.12	0.66	22.89	0.00	0.00
BK-90-3	P	23.67	2.61	0.16	2.23	0.33	19.40	0.00	0.00
GK-90-1	P	17.75	1.96	0.70	1.60	0.48	1.32	15.70	15.70
GK-90-2	P	19.27	2.13	0.27	2.17	0.72	11.97	17.90	17.90
GK-90-3	P	17.49	1.93	0.62	1.45	0.48	0.00	19.30	19.30
<i>Specimens conditioned in ocean water: specimens S</i>									
BS-30-1	P	29.39	3.26	0.01	2.33	0.85	29.39	0	0
BS-30-2	P	20.12	2.23	0.15	1.55	2.17	9.64	0	0
BS-30-3	P	23.57	2.61	0.12	2.07	0.31	N/A	0	0
GS-30-1	P	17.94	1.99	0.25	1.98	0.00	8.21	17.50	17.50
GS-30-2	P	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
GS-30-3	P	18.98	2.10	0.40	1.96	0.23	N/A	18.25	18.25
BS-60-1	P	24.03	2.61	0.1	2.00	0.55	9.61	0	0
BS-60-2	P	27.23	2.96	0.34	1.74	0.76	N/A	0	0
BS-60-3	P	25.74	2.8	0.13	1.96	1.14	9.28	0	0
GS-60-1	P	17.70	1.92	0.33	0.96	0.12	0.63	17.40	17.40
GS-60-2	P	17.81	1.93	0.18	1.44	0.00	10.16	17.04	17.04
GS-60-3	P	19.17	2.08	0.17	0.69	0.87	6.28	16.90	16.90
BS-90-1	P	22.62	2.51	0.14	2.08	0.23	5.76	0.00	0.00
BS-90-2	P	23.05	2.56	0.12	1.84	0.96	5.71	0.00	0.00
BS-90-3	P	26.01	2.89	0.10	1.54	0.92	20.50	0.00	0.00
GS-90-1	P	16.44	1.82	0.64	1.47	0.32	0.33	16.10	16.10
GS-90-2	P	17.33	1.92	0.35	0.75	0.93	0.77	15.96	15.96
GS-90-3	P	19.24	2.14	0.43	1.17	0.23	1.89	17.10	17.10

$\tau_{\max}$ = maximum bond stress; $\tau_{\max}/\sqrt{f'_c}$ = normalized bond stress; Suffix le and fe refers to loaded end and free (unloaded) end respectively; $s_{\max, le}$ and $s_{\max, fe}$ = slip corresponding to maximum stress; $\tau_{\text{ons}, le}$ and $\tau_{\text{ons}, fe}$ = bond stress at onset of slip; $\tau_{r, le}$ and $\tau_{r, fe}$ = residual bond stress; P = pullout; S = split; N/A = test data not available.

Table 4
Concrete compressive strengths after conditioning.

Environmental exposure	Duration (days)	f_c (MPa)
Unconditioned	–	60.0
Acid solution	30	74.0
	60	64.5
	90	55.0
Ocean water	30	81.5
	60	84.8
	90	81.2
Alkaline solution	30	70.0
	60	84.1
	90	82.1

The post peak responses were different for both types of bars due to their different surface treatments. The unconditioned GC specimens exhibited a gradual decrease in their bond stress with a noticeable load hardening as the pullout force increased (Fig. 3c and d). On the other hand, a sharp drop in the bond stress accompanied with insignificant slip in the bar was encountered for all unconditioned BC specimens (Fig. 3a and b). The sudden decrease in the bond stress explains the abrupt mode of failure that was observed during the test of the BFRP specimens. Bond–slip curves of the conditioned BFRP and GFRP specimens showed similar trends, as shown in Fig. 4 and Fig. 5, except for those specimens that failed due to concrete splitting, as will be described later.

It can be noticed that the unconditioned specimens reinforced with BFRP bars showed higher bond strength (27.8 MPa) than those reinforced with GFRP bars (18.4 MPa). This was attributed

to the type of the surface treatment of the specimens. Previous studies confirmed that sand-coated bars had higher bond strength than the ribbed bars [3]. On the other hand, the unconditioned BFRP specimens exhibited an average slip of 31% and 46% more than the GFRP specimens at the loaded and unloaded ends, respectively. The smaller slip values recorded for GFRP specimens correspond to their small bond strength as compared to BFRP specimens.

3.3. Failure mechanism

Table 3 shows the mode of failure of all pullout specimens. All specimens were split after testing to observe the bar and concrete conditions in the embedment zone. The bar diameters were also measured to assess the bar degradation after conditioning.

Unconditioned BFRP bars failed by inter-laminar shear between the bar layers. This was indicated by the removal of the outer sand-coated layer along the embedded portion of the bar and by the reduction in the bar diameter (Fig. 6a). An average reduction in diameter of 6% of the initial nominal diameter of the BFRP bars was observed. Unconditioned GFRP specimens failed after the ribs crushed of the ribs on the outer surface of the bar. Traces of resin and fiber residues were clearly noticed on the concrete surface in the embedded zone as shown in Fig. 6b. It is important to note that the failure mechanism of both types of bars was consistent with the observed bond–slip relationships. The successive demolition of the ribs during the pullout of the GFRP bars resulted in a ductile mode of failure after the peak stress was attained. In case of BFRP

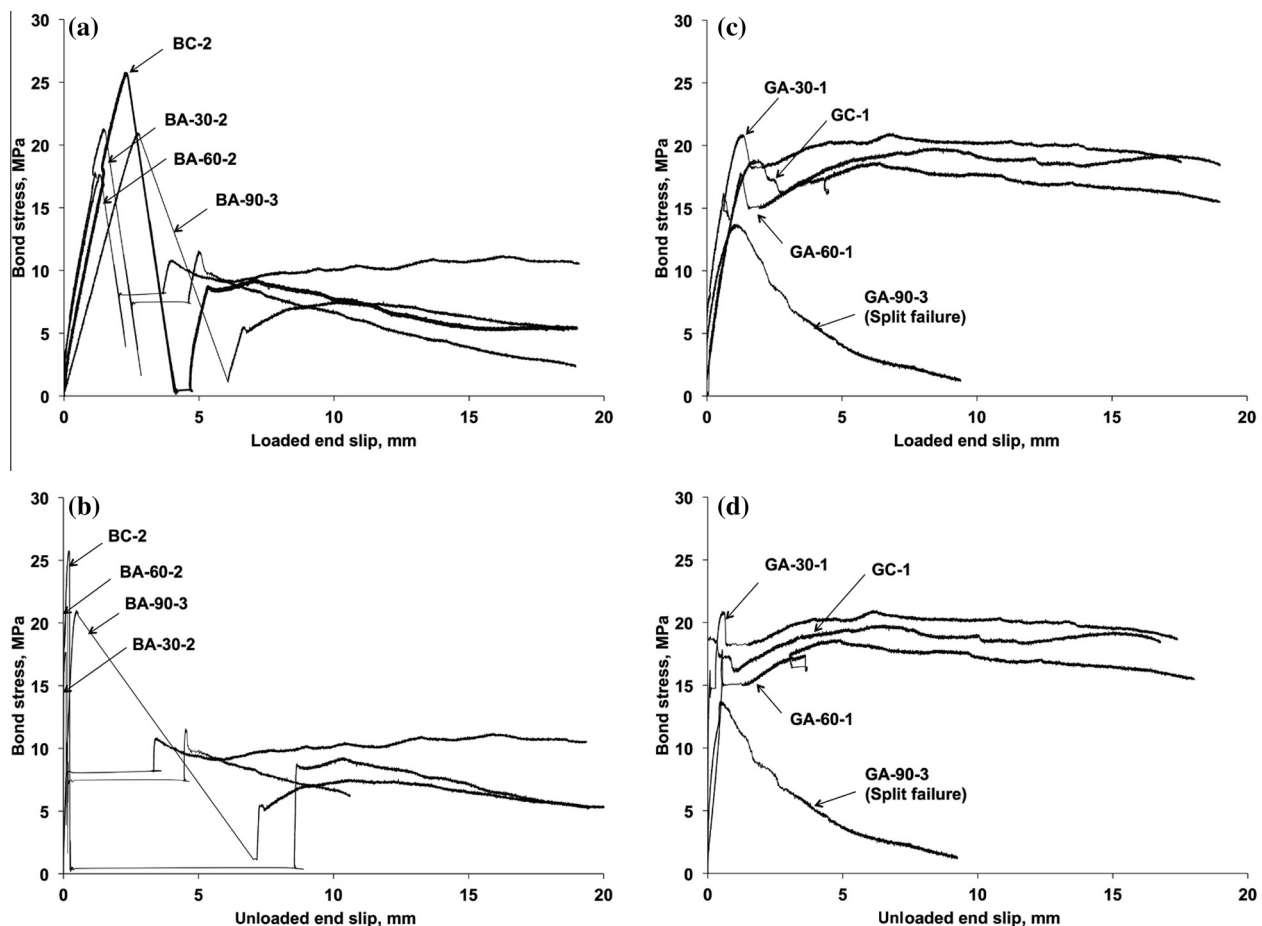


Fig. 3. Bond stress–slip curves at loaded and unloaded ends for (a and b) BFRP bars and (c and d) GFRP bars exposed to acid solutions.

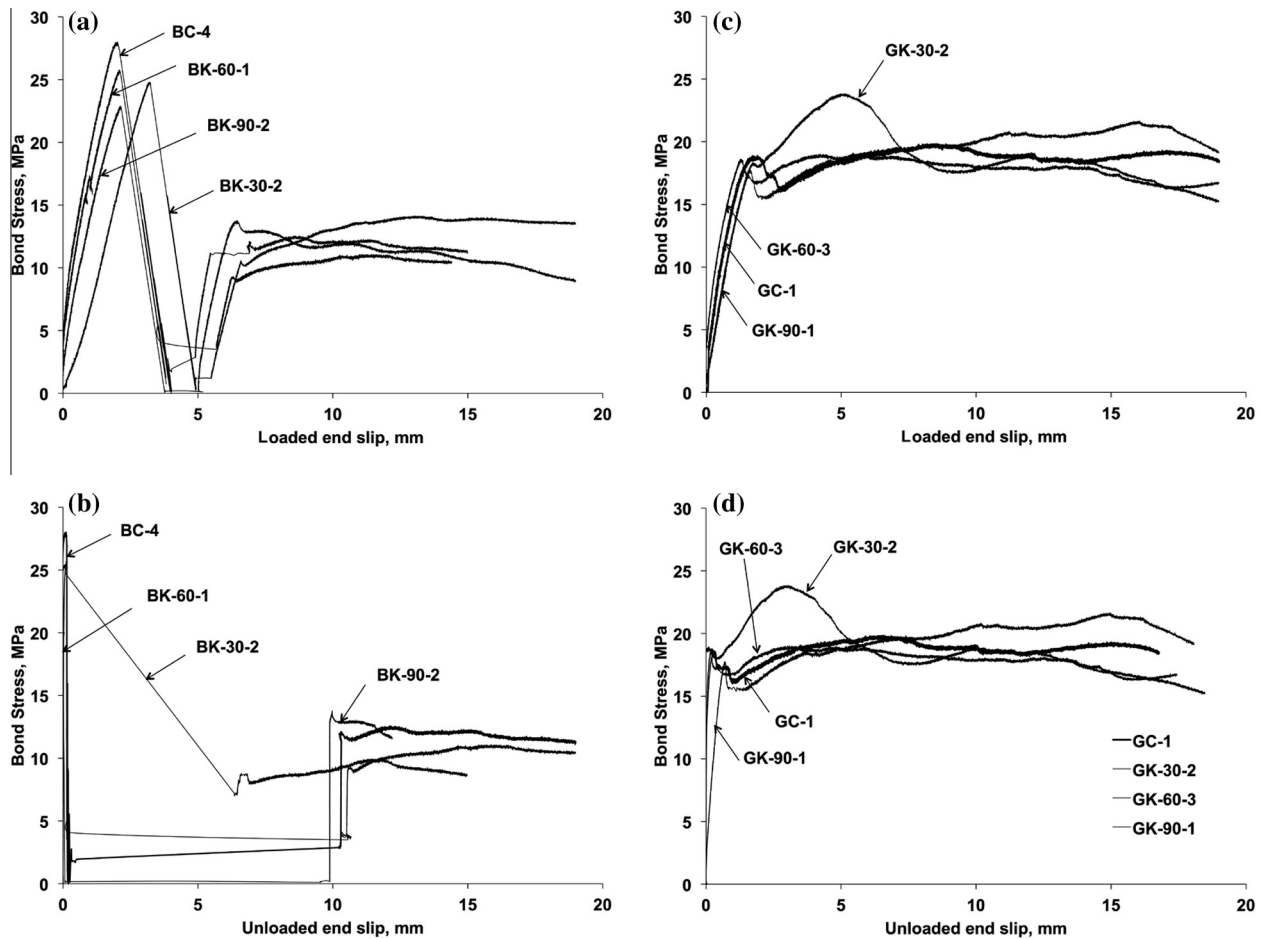


Fig. 4. Bond stress–slip curves at loaded and unloaded ends for (a and b) BFRP bars and (c and d) GFRP bars exposed to alkaline solutions.

bars, the sudden delamination of the sand layers explained the abrupt mode of failure of the bar at the end of the test.

Pullout mode of failure also characterized the conditioned BFRP specimens. For BFRP specimens immersed in acid and alkaline solutions, a brownish color was observed along the embedded portion of the bar and on the surrounding concrete surface. This finding denoted that a chemical reaction took place between the FRP material and the solution, which implied the diffusion of the solution molecules into the embedded zone through the concrete voids and pores. The brownish color was more pronounced in BFRP specimens that were immersed for 60 and 90 days (Fig. 7a). No color change was observed in the BFRP specimens embedded in ocean water. For all conditioned BFRP specimens, no visual deterioration of concrete was noticed.

On the other hand, GFRP specimens GA-30-2, GA-60-2, GA-90-1, and GA-90-3 failed by concrete splitting rather than by pullout of the bar. This mode of failure gave evidence to the detrimental effect of the acid solution on both concrete and GFRP bars. The GFRP bars in these specimens showed minor scratches on their embedded lengths while major portions of their ribs could still be observed on the bar surface (Fig. 7b). All other conditioned GFRP bars failed in typical pullout mode similar to that observed in their unconditioned counterparts. In these specimens, the ribs were completely eaten on the bar surface with white traces of resin and fiber residues being observed on the concrete surface at the location of the bar. For GFRP specimens immersed in alkaline solutions, no traces of chemical attack could be detected, whereas specimens immersed in ocean water for 90 days exhibited removal of the resin and the adjacent fiber layers.

4. Variation of bond strength in conditioned specimens

4.1. Bond strength

The plot shown in Fig. 8 compares the average bond strengths of both unconditioned and conditioned specimens. In order to eliminate the effect of variation of concrete compressive strength on the test results, normalized values with respect to the square root of concrete strength were determined and shown in Fig. 9.

Test results revealed that all conditioned specimens exhibited deterioration in their bond strength. For BFRP bars, acid and ocean water had the most detrimental effect on their bond strength after 30 days of exposure. The reduction in the normalized bond strength reached 25% of the bond strength of the unconditioned specimens compared to 14% loss in bond for specimens immersed in the alkaline solution. After 60 and 90 days of immersion, the bond strength of BFRP bars slightly increased or remained unchanged in acid and ocean water solutions. This can be depicted in the plots of Fig. 9. The increase in bond strength of BFRP bars after long exposure in acid solution can be attributed to the swelling of the bar during conditioning, which increased the friction between the bars and the surrounding concrete. This finding was supported by the high moisture absorption reported in the manufacturers' data sheets for the BFRP bars (0.45%). BFRP specimens immersed in alkaline solutions suffered a continuous loss in bond strength that reached 19% and 25% after 60 and 90 days of immersion, respectively. The reduction in bond strength indicated the reactivity of the bar material with the alkaline solution following the diffusion of the solution through the concrete voids. On

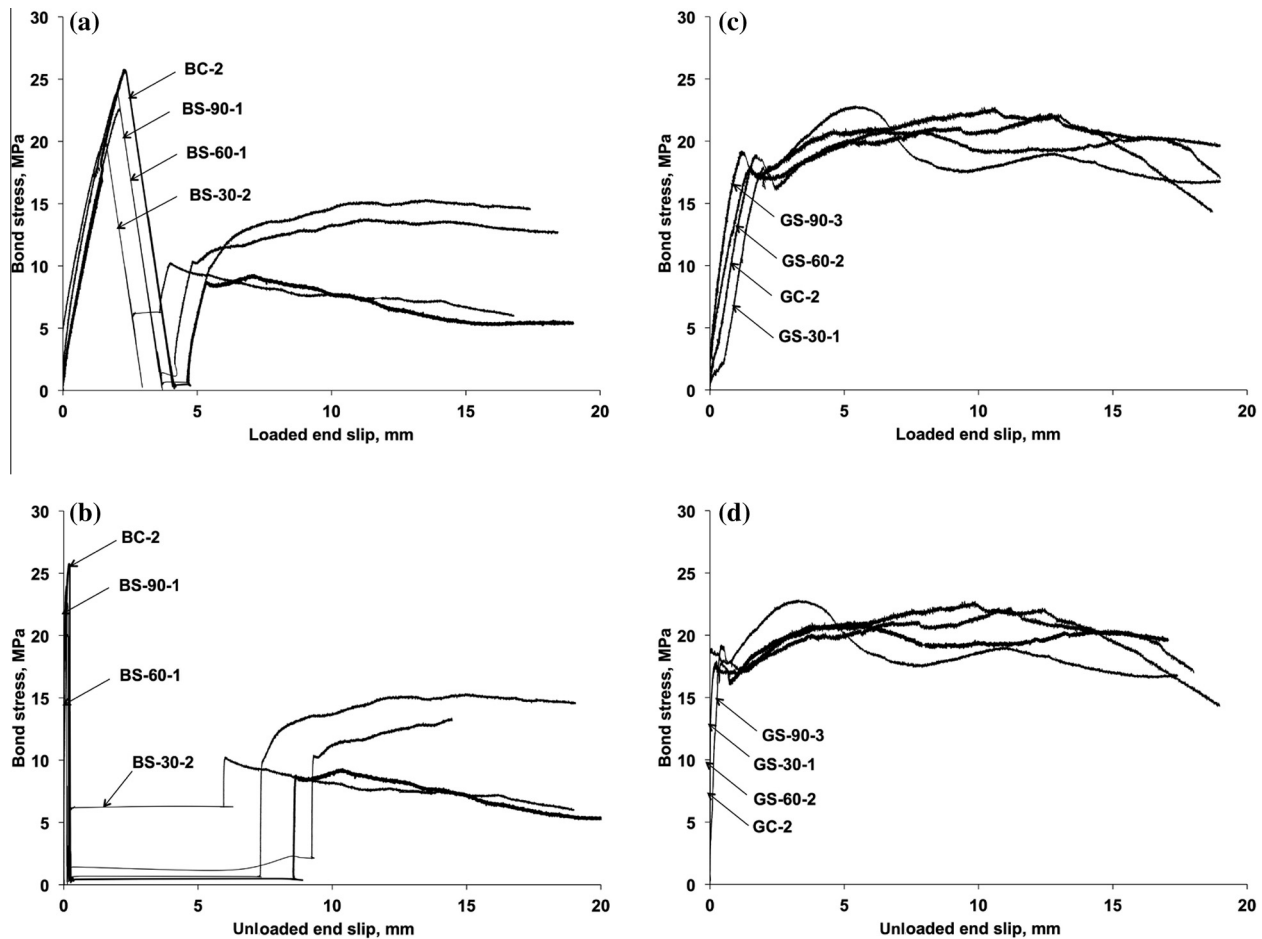


Fig. 5. Bond stress-slip curves at loaded and unloaded ends for (a and b) BFRP bars and (c and d) GFRP bars exposed to ocean water.

the other hand, GFRP specimens immersed in acid solution suffered a bond loss of 25% after 90 days, compared to 17% loss in strength after immersion in both ocean water and alkaline solutions.

4.2. Slip and stiffness of the FRP bar

Fig. 10 shows the average slip, s_{max} , recorded at maximum stress at the loaded and unloaded ends. Insignificant slip was recorded at the unloaded ends for all specimens. At the loaded ends, the loss of bond strength in the BFRP bars in acid and ocean water solutions was accompanied by a decrease in the slip recorded at the loaded ends of the bars. The variation in slip after long 90 days of immersion was negligible as can be seen in Fig. 10a. Slip measurements were consistent with the obtained bond strengths. The stiffness of the BFRP specimens did not vary significantly after conditioning, as shown from the insignificant variation in the slope of the bond-slip curves depicted in Fig. 3 and Fig. 5. This finding indicated a stable behavior of the BFRP bars in acid and ocean water exposures. On the other hand, BFRP specimens immersed in alkaline solutions for 30 days suffered an increase in slip of 21% of the slip encountered for the unconditioned specimens at the loaded ends. The slip then decreased after 60 and 90 days of continuous alkaline exposure. The consistent variation in slip measurements with the maximum stress attained indicated that the stiffness of the bars had been altered after being immersed in the alkaline solution. In fact, BFRP bars lost about 14% of their stiffness after 90 days of alkaline exposure. Loss of stiffness was noticed from the change in slope of the bond-slip curves shown in Fig. 4b. This finding supported the fact that chemical reaction took

place between the alkaline solution and the BFRP bars. On the other hand, conditioned GFRP bars encountered a decrease in slip after conditioning, which was also consistent with the encountered decrease in the bond strength of the bars. However, discrepancy in the recorded slip of GFRP specimens indicated the random effect of conditioning on the behavior of the bars.

4.3. Adhesion stresses

Adhesion is the principal bond component that describes the bond of FRP bars to concrete at initial stages of loading. The stress at the onset of slip at the loaded end is the stress at which the adhesion breaks between the bar and the surrounding concrete. Fig. 11 compares between the average onset stresses, τ_{onset} , for both unconditioned and conditioned specimens. These stresses were determined by visual inspection of the bond-slip curves for all specimens at the points where the slope of the ascending branch changed abruptly indicating the breakage of adhesion between the bar and concrete.

The adhesion strengths of BFRP and GFRP unconditioned specimens were 0.52 MPa and 0.41 MPa, respectively. Conditioning of specimens caused noticeable variation in their adhesion strength, particularly at early stages of immersion as illustrated in Fig. 11. All BFRP and GFRP conditioned specimens exhibited an increase in adhesion stress after 30 days of exposure. The only exception was for the GFRP specimens immersed in ocean water. These specimens exhibited a decrease in their adhesion stress after 30 days of immersion.

The increase in adhesion stresses at early stages of conditioning was attributed to the swelling of the bar, as previously noted, and

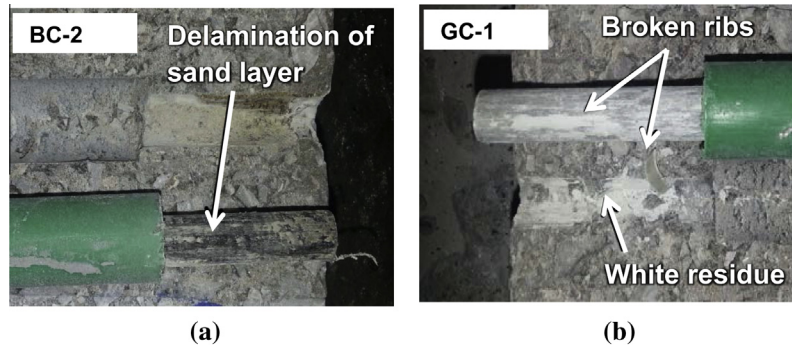


Fig. 6. Pullout failure of unconditioned (a) BFRP and (b) GFRP bars.

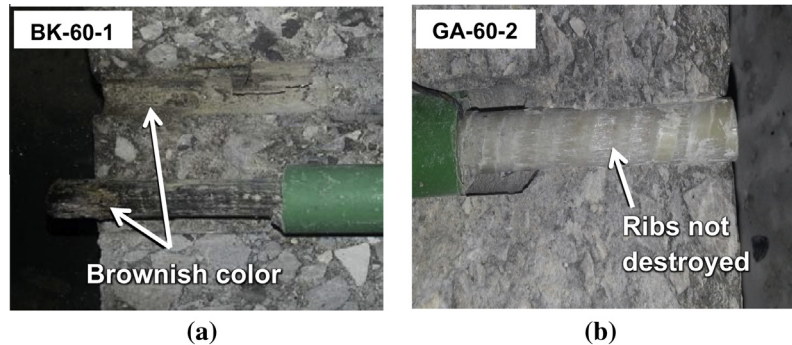


Fig. 7. Pullout failure for conditioned (a) BFRP and (b) GFRP specimens.

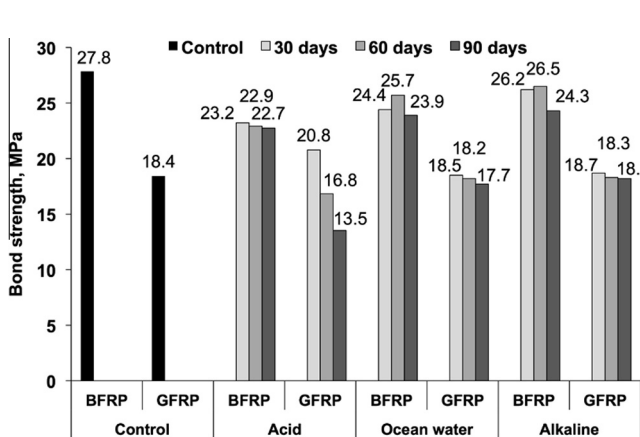


Fig. 8. Variation of bond strength for BFRP and GFRP specimens.

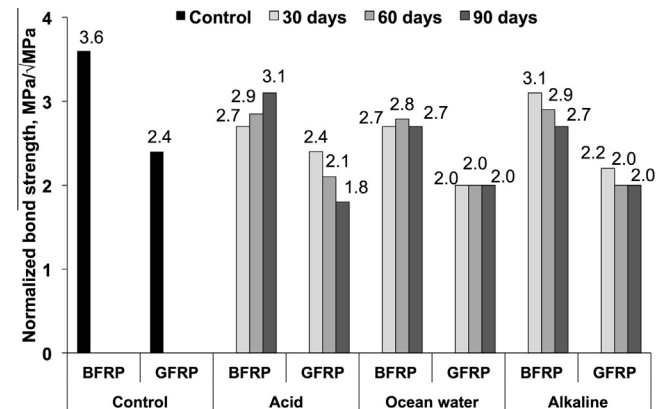


Fig. 9. Normalized bond strength for BFRP and GFRP specimens.

to the increase in the concrete strength at this stage. This increase was more pronounced in BFRP than in GFRP conditioned specimens due to the high absorption properties of BFRP bars (0.45% compared to 0.024% for GFRP bars). For long duration of exposure, the rate of dissipation of the solution molecules in the FRP material and the concrete voids diminished, which concluded the increase in the bar diameter and consequently, no further increase in adhesion stresses was observed. On the contrary, the gradual deterioration of the surface coating of the FRP bars counteracted the swelling effect and caused the deterioration of adhesion to concrete. This was more evident in the BFRP specimens immersed in the acid solution. The reactivity of BFRP bars with the acid solution was dominant in deteriorating the adhesion stress. On the other hand, conditioned GFRP specimens exhibited similar behavior with less increase in adhesion stresses than the BFRP specimens after

30 days of immersion. Discrepancies in the adhesion stresses were noticed in the GFRP specimens immersed in ocean water probably due to the random deterioration of the bar. Minor changes in adhesion stresses were encountered for GFRP bars in the alkaline solution.

It is important to note that unconditioned and conditioned BFRP specimens exhibited higher adhesion strengths than their counterparts GFRP specimens, which can be explained by the difference in the characteristics of their interfacial surface in contact with concrete. Sand-coated bars usually exhibit higher adhesion strength than ribbed-surface bars [18].

5. Theoretical approach

The variation in the bond-slip behavior of conditioned BFRP and GFRP bars necessitates the calibration of the existing bond models

to best describe such behavior. Eligehausen et al. [14] developed an analytical model (the BPE model) to depict the bond-slip relationship for steel rods. Previous studies calibrated the model for GFRP and CFRP bars [9] but without considering the effect of environmental exposures on their bond performance. The model described the ascending branch of the bond-slip curve using the equation below:

$$\frac{\tau}{\tau_{\max}} = \left(\frac{s}{s_{\max}} \right)^{\alpha}$$

where τ and s are the bond stress and the corresponding slip at any stage of load application, respectively, while $\tau_{\max}$ and $s_{\max}$ are their corresponding maximum values. The parameter α is a curve fitting parameter that was calibrated for the unconditioned and conditioned BFRP and GFRP bars from the current test results. This parameter depends on the characteristics of the FRP bar, its surface treatment, and the exposure conditions. For the indented, grained covered, and ribbed FRP bars, Cosenza et al. [9] reported average values for α as 0.177, 0.067, and 0.283, respectively.

In the current study, the parameter α was evaluated by equating the areas underneath the ascending branches of the analytical and experimental bond-slip curves at the unloaded ends [8,11,12]. Good correlation between the experimental and analytical data can be depicted in Fig. 12 for representative BFRP and GFRP specimens. Table 5 shows the calibrated average values of α for each conditioning exposure as determined from the current test results. Values of α for unconditioned specimens are also shown.

From the test results, a value of $\alpha = 0.015$ was recommended for the unconditioned BFRP and GFRP specimens to describe the practically linear behavior of both types of bars and the minimal slip that occurred in the bars close to the maximum stress. For BFRP

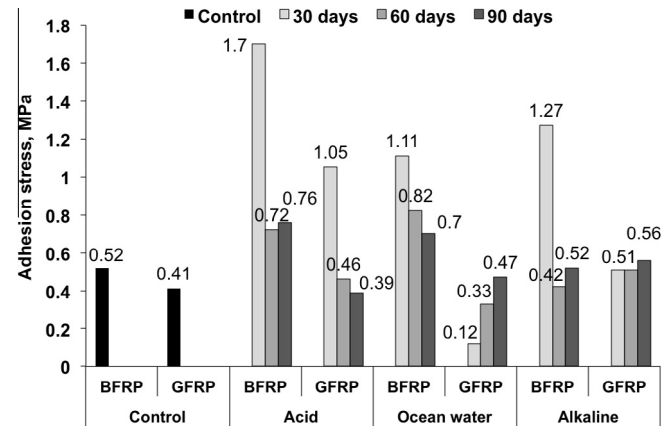


Fig. 11. Adhesion stress for BFRP and GFRP specimens.

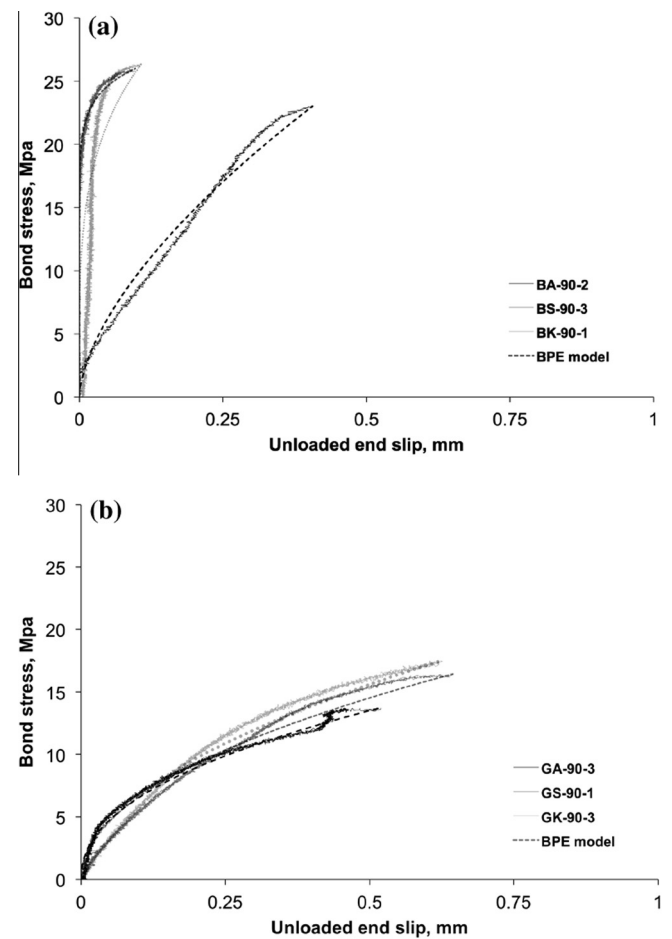


Fig. 12. Comparison between the analytical and experimental ascending branches of the bond stress-slip curves for (a) BFRP and (b) GFRP specimens immersed for 90 days.

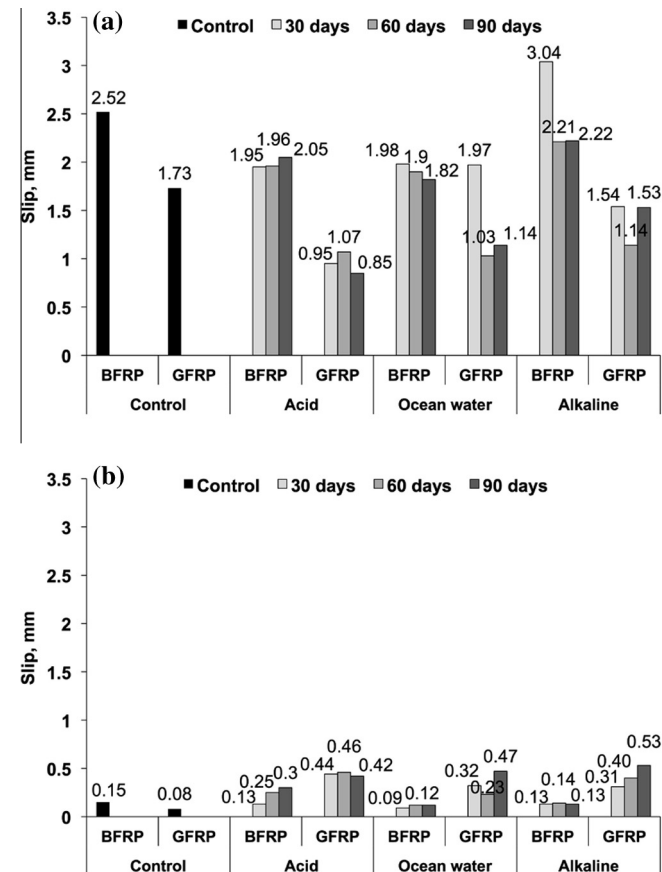


Fig. 10. Slip at maximum stress at the (a) loaded and (b) unloaded ends.

conditioned specimens, the most detrimental effect on the bond-slip behavior was encountered after 90 days of acid immersion. For these specimens, the calibrated parameter α increased to 0.65 compared to 0.56 for ribbed GFRP specimens conditioned in alkaline solution for 90 days. Based on these results, safe design values for α are recommended as 0.25 and 0.56 for both sand-coated BFRP and ribbed GFRP, respectively. An exception is noted for BFRP bars that might be exposed to acid environments; a value of $\alpha = 0.65$ should therefore be considered.

Table 5Mean values of parameter α for different environmental exposures.

Type of exposure	α_{BFRP}	α_{GFRP}
Unconditioned	0.015	0.015
Acid solution – 30 days	0.100	0.185
Acid solution – 60 days	0.095	0.265
Acid solution – 90 days	0.650	0.450
Ocean water – 30 days	0.150	0.180
Ocean water – 60 days	0.180	0.193
Ocean water – 90 days	0.080	0.530
Alkaline solution – 30 days	0.200	0.190
Alkaline solution – 60 days	0.175	0.150
Alkaline solution – 90 days	0.250	0.560

6. Conclusions and recommendations

Sixty-two sand-coated BFRP and ribbed GFRP unconditioned and conditioned specimens were tested under direct pullout tests. The following conclusions were drawn from the study:

- Conditioning of pullout specimens in aqueous solutions had negligible effect on the trend of the bond stress–slip curves.
- Sand-coated bars showed higher bond strength and higher adhesion to concrete than ribbed bars regardless of the fiber type. This finding was observed for both unconditioned and conditioned specimens.
- Swelling of the bar at early stages of exposure enhanced its adhesion to concrete. This phenomenon depends on the absorption properties of the FRP material and the quality of the manufacturing of the bar. Continuous immersion in solutions caused rapid deterioration in adhesion stresses.
- The inter-laminar shear between the FRP layers governed the failure of both unconditioned and conditioned pullout specimens rather than the shear stress between the bar and concrete.
- BFRP specimens immersed in ocean water and alkaline solution for 90 days exhibited 25% reduction in their bond strength compared to 14% for BFRP specimens immersed in acid solution.
- GFRP specimens exposed to acid solution for 90 days suffered the highest loss in bond strength (25%) compared to 17% for specimens immersed in alkaline and ocean water.
- Split failure rather than pullout was the governing mode of failure that characterized most of GFRP specimens conditioned in acid solutions.
- The variations in slip of BFRP and GFRP bars after conditioning were negligible. However, BFRP bars lost average of 14% of their initial stiffness due to alkaline exposure for 90 days.

The present study confirmed the great potential of using BFRP bars as reinforcing materials. Further research should be devoted to include other commercially available bars. It is important to mention that all specimens were directly exposed to saturated solutions until testing, which might not represent the actual condi-

tions of FRP bars in service. Furthermore, the experimental work presented is limited to the bars used in this study and should not be extended to other bar types.

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